

```
import numpy as np
import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
fake = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/dataset_fake_true/Fake.csv')
```

```
true = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/dataset_fake_true/True.csv')
```

```
true.head()
```

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017

```
fake.head()
```

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
true['label']=1
```

```
fake['label']=0
```

```
true.head()
```

	title	text	subject	date	label
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017	1
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017	1
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017	1

Data Preprocessing

```
news = pd.concat([true,fake],axis=0)
```

```
news.head()
```

	title	text	subject	date	label
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017	1
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017	1
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017	1

```
news.tail()
```

		title	text	subject	date	label
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	0	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-ac...	21st Century Wire says It s a familiar theme. ...	Middle-east	January 16, 2016	0	
23478	Sunnistan: US and Allied 'Safe Zone' Plan to T...	Patrick Henningsen 21st Century WireRemember ...	Middle-east	January 15, 2016	0	

```
news.isnull().sum()
```

	0
title	0
text	0
subject	0
date	0
label	0

dtype: int64

```
news = news.drop(['title', 'subject', 'date'],axis=1)
```

```
news.head()
```

		text	label
0	WASHINGTON (Reuters) - The head of a conservat...		1
1	WASHINGTON (Reuters) - Transgender people will...		1
2	WASHINGTON (Reuters) - The special counsel inv...		1
3	WASHINGTON (Reuters) - Trump campaign adviser ...		1
4	SEATTLE/WASHINGTON (Reuters) - President Donal...		1

```
news=news.sample(frac=1)
```

```
news.head()
```

		text	label
5980	PHILADELPHIA (Reuters) - President Donald Trum...		1
17656	LONDON (Reuters) - Sally Jones, a British jiha...		1
10829	MARFA, Texas (Reuters) - U.S. Supreme Court Ju...		1
19285	Hundreds of protestors and LGBTQ activists too...		0
2873	HAMBURG (Reuters) - French President Emmanuel ...		1

```
news.reset_index(inplace=True)
```

```
news.head()
```

	index		text	label
0	5980	PHILADELPHIA (Reuters) - President Donald Trum...		1
1	17656	LONDON (Reuters) - Sally Jones, a British jiha...		1
2	10829	MARFA, Texas (Reuters) - U.S. Supreme Court Ju...		1
3	19285	Hundreds of protestors and LGBTQ activists too...		0
4	2873	HAMBURG (Reuters) - French President Emmanuel ...		1

```
news.drop(['index'],axis=1,inplace=True)
```

```
news.head()
```

	text	label
0	PHILADELPHIA (Reuters) - President Donald Trum...	1
1	LONDON (Reuters) - Sally Jones, a British jiha...	1
2	MARFA, Texas (Reuters) - U.S. Supreme Court Ju...	1
3	Hundreds of protestors and LGBTQ activists too...	0
4	HAMBURG (Reuters) - French President Emmanuel ...	1

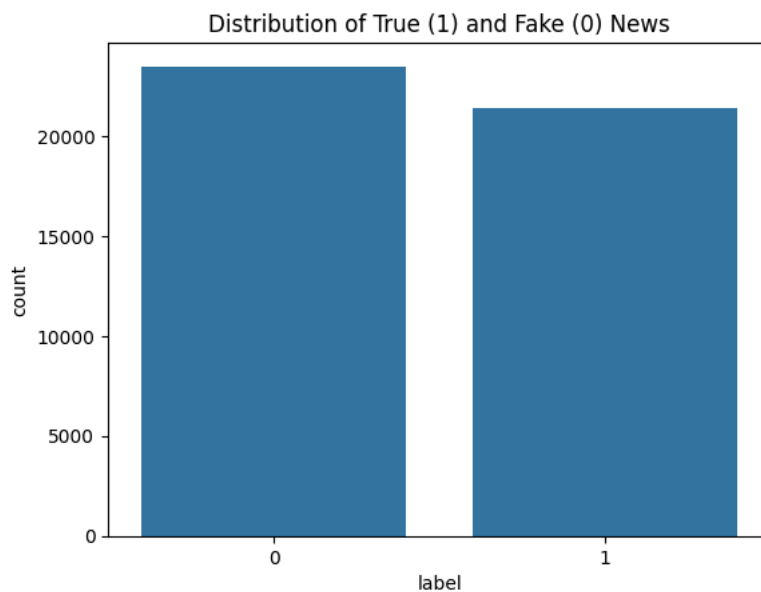
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load datasets
fake = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/dataset_fake_true/Fake.csv")
true = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/dataset_fake_true/True.csv")

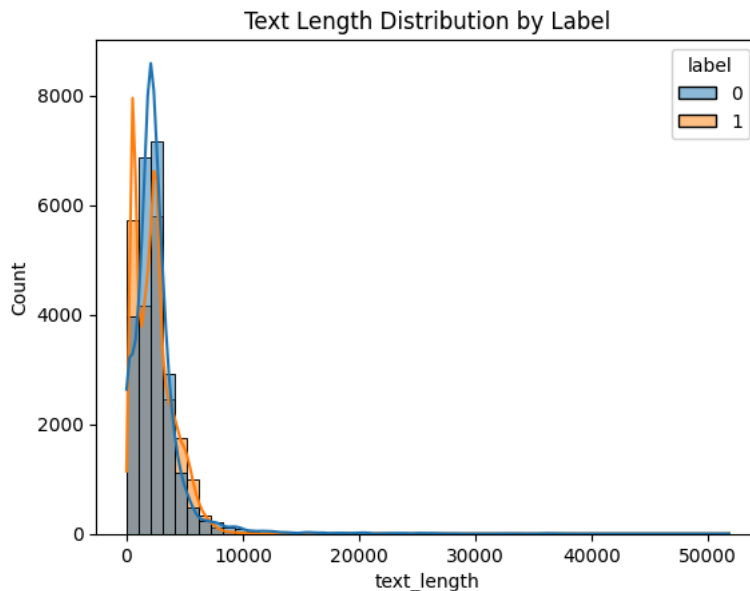
# Add labels: 0 for fake, 1 for true
fake['label'] = 0
true['label'] = 1

# Combine both datasets
data = pd.concat([fake, true], axis=0).reset_index(drop=True)

# Visualize the distribution
sns.countplot(x='label', data=data)
plt.title("Distribution of True (1) and Fake (0) News")
plt.show()
```



```
#Analyze Text Length
data['text_length'] = data['text'].apply(lambda x: len(str(x)))
sns.histplot(data=data, x='text_length', bins=50, hue='label', kde=True)
plt.title("Text Length Distribution by Label")
plt.show()
```



```
import re
from nltk.corpus import stopwords
```

```
import nltk

# Step 1: Download the stopwords dataset
nltk.download('stopwords')

# Step 2: Import stopwords from the corpus
from nltk.corpus import stopwords

# Step 3: Use stopwords
stop_words = stopwords.words('english')

print(stop_words[:10])
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

```
# Import necessary libraries
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

# Make sure stopwords are downloaded
nltk.download('stopwords')
import re
import string

def wordopt(text):
    text = text.lower()
    text = re.sub(r'[\.\*\?]', '', text) # remove text inside []
    text = re.sub(r'\"\\w\", \" \", text) # remove non-word characters
    text = re.sub(r'https?://\S+|www\.\S+', '', text) # remove URLs
    text = re.sub(r'<.*?>+', '', text) # remove HTML tags
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text) # remove punctuation
    text = re.sub(r'\n', ' ', text) # remove newlines
    text = re.sub(r'\w*\d\w*', '', text) # remove words with numbers
    return text
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```

```
def wordopt(text):
    #convert to lower case
    text = text.lower()

    #remove punctuation
    text = re.sub('[^a-zA-Z]', ' ', text)

    #remove URLs
    text = re.sub('https?://\S+|www\.\S+', '', text)
```

```
#remove stopwords
text = text.split()
text = [word for word in text if not word in stopwords.words('english')]

#remove digits
text = [word for word in text if not word.isdigit()]

#stemming
ps = PorterStemmer()
text = [ps.stem(word) for word in text]

#remove newline characters
text = [word.replace('\n','') for word in text]

#join the words
text = ' '.join(text)

return text
```

```
<>:9: SyntaxWarning: invalid escape sequence '\S'
<>:9: SyntaxWarning: invalid escape sequence '\S'
/tmp/ipython-input-2394798696.py:9: SyntaxWarning: invalid escape sequence '\S'
text = re.sub('https?://\S+|www\.\S+', '', text)
```

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```
news['text']
```

	text
0	PHILADELPHIA (Reuters) - President Donald Trum...
1	LONDON (Reuters) - Sally Jones, a British jiha...
2	MARFA, Texas (Reuters) - U.S. Supreme Court Ju...
3	Hundreds of protestors and LGBTQ activists too...
4	HAMBURG (Reuters) - French President Emmanuel ...
...	...
44893	BERLIN (Reuters) - German Chancellor Angela Me...
44894	Remember the criticism George W. Bush received...
44895	MADRID (Reuters) - The Spanish government has ...
44896	BEIRUT (Reuters) - Iranian Foreign Minister Mo...
44897	

44898 rows × 1 columns

dtype: object

```
from re import X
x = news['text']
y= news['label']
```

x

	text
0	PHILADELPHIA (Reuters) - President Donald Trum...
1	LONDON (Reuters) - Sally Jones, a British jiha...
2	MARFA, Texas (Reuters) - U.S. Supreme Court Ju...
3	Hundreds of protestors and LGBTQ activists too...
4	HAMBURG (Reuters) - French President Emmanuel ...
...	...
44893	BERLIN (Reuters) - German Chancellor Angela Me...
44894	Remember the criticism George W. Bush received...
44895	MADRID (Reuters) - The Spanish government has ...
44896	BEIRUT (Reuters) - Iranian Foreign Minister Mo...
44897	

44898 rows × 1 columns

dtype: object

y

	label
0	1
1	1
2	1
3	0
4	1
...	...
44893	1
44894	0
44895	1
44896	1
44897	0

44898 rows × 1 columns

dtype: int64

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.3)
```

```
x_train.shape
```

(31428,)

```
x_test.shape
```

(13470,)

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
vectorization = TfidfVectorizer()
```

```
xv_train = vectorization.fit_transform(x_train)
```

```
xv_test = vectorization.transform(x_test)
```

```
xv_train
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
  with 6558103 stored elements and shape (31428, 105640)>
```

```
xv_test
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
  with 2820596 stored elements and shape (13470, 105640)>
```

ml model logistic regression

```
from sklearn.linear_model import LogisticRegression
```

```
LR = LogisticRegression()
```

```
LR.fit(xv_train,y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression()
```

```
LR.score(xv_test,y_test)
```

```
0.9878990348923534
```

```
pred_lr = LR.predict(xv_test)
```

```
from sklearn.metrics import classification_report
```

```
print(classification_report(y_test,pred_lr))
```

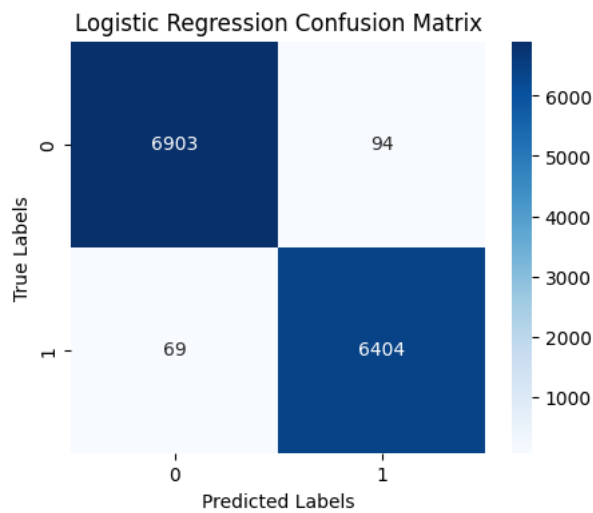
	precision	recall	f1-score	support
0	0.99	0.99	0.99	6997
1	0.99	0.99	0.99	6473
accuracy			0.99	13470
macro avg	0.99	0.99	0.99	13470
weighted avg	0.99	0.99	0.99	13470

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
import seaborn as sns
import matplotlib.pyplot as plt
```

```
cm = confusion_matrix(y_test, pred_lr)
print("\nConfusion Matrix:\n", cm)
```

```
Confusion Matrix:
[[6903  94]
 [ 69 6404]]
```

```
plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted Labels')
plt.ylabel('True Labels')
plt.title('Logistic Regression Confusion Matrix')
plt.show()
```



decision tree *classifier*

```
from sklearn.tree import DecisionTreeClassifier
```

```
DT = DecisionTreeClassifier()
```

```
DT.fit(xv_train,y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier()
```

```
DT.score(xv_test,y_test)
```

```
0.9957683741648107
```

```
pred_dt = DT.predict(xv_test)
```

```
print(classification_report(y_test,pred_dt))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	6997
1	1.00	1.00	1.00	6473
accuracy			1.00	13470
macro avg	1.00	1.00	1.00	13470
weighted avg	1.00	1.00	1.00	13470

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Train Decision Tree model
dt = DecisionTreeClassifier(random_state=42)
dt.fit(xv_train, y_train)

# Make predictions
pred_dt = dt.predict(xv_test)

# Create confusion matrix
cm = confusion_matrix(y_test, pred_dt)

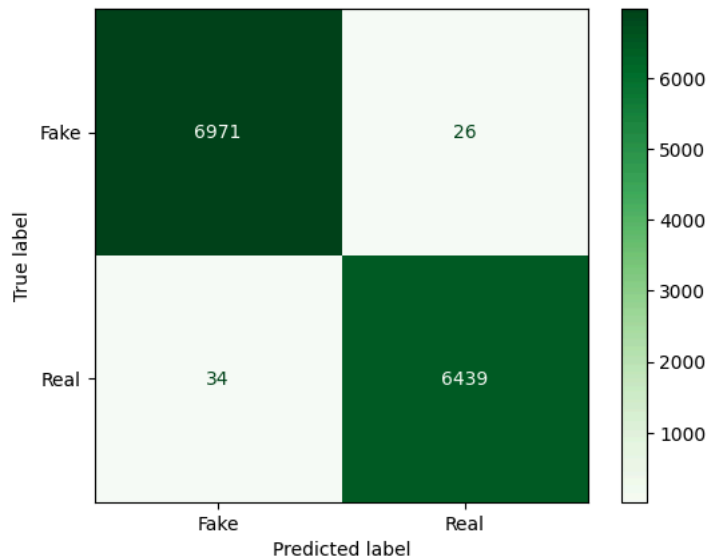
# Print numeric confusion matrix
print("Confusion Matrix:\n", cm)

# Display confusion matrix as a heatmap
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Fake", "Real"])
disp.plot(cmap='Greens')
plt.title("Decision Tree Confusion Matrix - Fake News Prediction")
plt.show()
```

Confusion Matrix:

```
[[6971  26]
 [ 34 6439]]
```

Decision Tree Confusion Matrix - Fake News Prediction



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random forest

```
from sklearn.ensemble import RandomForestClassifier
```

```
RF = RandomForestClassifier(random_state=0)
```

```
RF.fit(xv_train,y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=0)
```

```
RF.score(xv_test,y_test)
```

```
0.9907943578322197
```

```
pred_rf = RF.predict(xv_test)
```

```
print(classification_report(y_test,pred_rf))
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	6997
1	0.99	0.99	0.99	6473
accuracy			0.99	13470
macro avg	0.99	0.99	0.99	13470
weighted avg	0.99	0.99	0.99	13470

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

```
# Train Random Forest model
rf = RandomForestClassifier(random_state=42)
rf.fit(xv_train, y_train)
```

```
# Make predictions
pred_rf = rf.predict(xv_test)
```

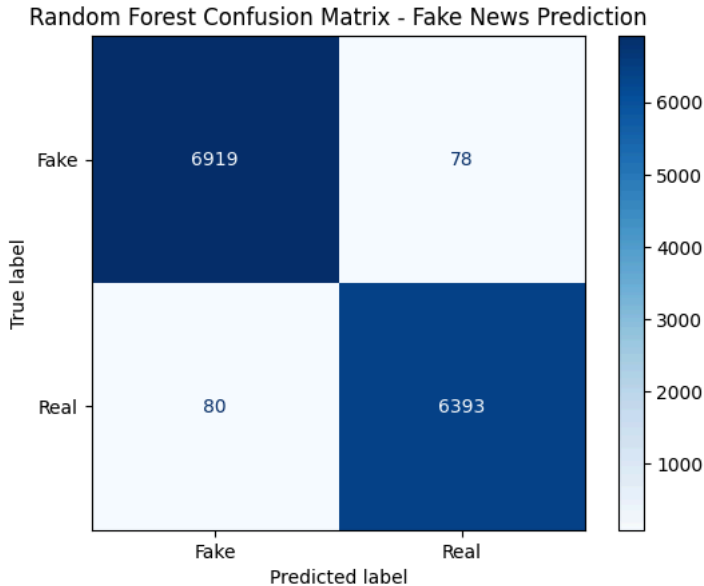
```
# Create confusion matrix
cm = confusion_matrix(y_test, pred_rf)
```

```
# Print numeric confusion matrix
print("Confusion Matrix:\n", cm)

# Display confusion matrix as a heatmap
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Fake", "Real"])
disp.plot(cmap='Blues')
plt.title("Random Forest Confusion Matrix - Fake News Prediction")
plt.show()
```

Confusion Matrix:

```
[[6919  78]
 [ 80 6393]]
```



predictive model

```
def output_label(n):
    if n == 0:
        return "Fake News"
    elif n == 1:
        return "True News"
```

Start coding or [generate](#) with AI.

```
def manual_testing(news):
    # build DataFrame
    testing_news = {"text": [news]}
    new_def_test = pd.DataFrame(testing_news)

    # preprocess (wordopt must be defined)
    new_def_test["text"] = new_def_test["text"].apply(wordopt)
    new_x_test = new_def_test["text"]

    # vectorize (vectorization must be fitted already)
    new_xv_test = vectorization.transform(new_x_test)

    # predict (take first element because we passed a single sample)
    pred_LR = LR.predict(new_xv_test)[0]
    pred_RF = RF.predict(new_xv_test)[0]

    # try DT if available
    try:
        pred_DT = DT.predict(new_xv_test)[0]
    except NameError:
        pred_DT = None
    except Exception:
        pred_DT = None

    # safe label conversion (fix name if your function is output_label)
    def safe_label(p):
        if p is None:
            return "N/A"
        return output_label(p) # replace with output_label if that's the actual name

    lr_label = safe_label(pred_LR)
    dt_label = safe_label(pred_DT)
```

```
rf_label = safe_label(pred_RF)

# print and return structured result (don't return print())
output_str = f"\nLR Prediction: {lr_label}\nDT Prediction: {dt_label}\nRF Prediction: {rf_label}"
print(output_str)

# also return a dict so caller can use values programmatically
return {"LR": lr_label, "DT": dt_label, "RF": rf_label}
```

```
#news = str(input())
#manual_testing(news)
```

```
def output_label(n):
    """Convert numeric prediction to readable label."""
    return "Fake News" if n == 0 else "True News"

def manual_testing(news):
    # Create DataFrame for the input text
    testing_news = {"text": [news]}
    new_def_test = pd.DataFrame(testing_news)

    # Preprocess text using your function
    new_def_test["text"] = new_def_test["text"].apply(wordopt)

    # Transform using your fitted vectorizer
    new_x_test = new_def_test["text"]
    new_xv_test = vectorization.transform(new_x_test)

    # Predict using your trained models
    pred_LR = LR.predict(new_xv_test)
```